
Training Protocol Dominates Rollout Drift in Latent World Models: A Causal Attribution Study

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Abstract

1 Latent world models suffer from rollout drift: small perturbations to the initial
2 latent state amplify exponentially during multi-step imagination. The dominant
3 explanation blames marginal encoder regularizers (e.g., SIGReg) for failing to con-
4 strain the conditional dynamics. We run a controlled $2 \times 2 \times 2$ factorial experiment
5 across three synthetic environments (stable-linear, 2D spiral, Lorenz) to causally
6 attribute rollout amplification (A_k) to three factors: encoder regularization (F1),
7 dynamics model capacity (F2), and training protocol (F3: 1-step teacher-forcing
8 vs. 5-step BPTT). Variance decomposition of $\log A_{50}$ shows F3 is the dominant
9 factor in two of three environments ($\eta^2 = 0.64$ on stable-linear, 0.44 on Lorenz),
10 with F1 second on Lorenz ($\eta^2 = 0.28$). Critically, we find a *sanity-check failure*:
11 all models trained with 1-step teacher-forcing exhibit $A_{50} \gg 1$ even on a trivially
12 stable linear system with eigenvalues $\{0.7, 0.8, 0.9\}$, where perturbations must
13 geometrically shrink. This reveals that the encoder learns geometrically unstable
14 latent representations regardless of regularizer, and that multi-step BPTT training
15 is the primary corrective mechanism. The “regularizer cannot bound rollout drift”
16 framing is technically correct but practically secondary: fixing the training protocol
17 matters more.

18 1 Introduction

19 Latent world models [Hafner et al., 2023, LeCun, 2022] compress observations into a latent space $\mathbf{z}$
20 and learn a dynamics function f such that $\mathbf{z}_{t+1} \approx f(\mathbf{z}_t, \mathbf{a}_t)$. A central failure mode is *rollout drift*:
21 errors compound during multi-step imagination, making long-horizon planning unreliable.

22 A popular diagnosis [Anonymous, 2025] attributes drift to marginal regularizers such as SIGReg,
23 which constrain the distribution $q(\mathbf{z})$ aggregated over timesteps but place no constraint on the
24 conditional $p(\mathbf{z}_{t+1}|\mathbf{z}_t)$. The implicit claim is: *fix the regularizer and rollout drift improves*.

25 We test this claim directly via causal attribution. We ask: of the total rollout amplification A_k
26 observed in latent world models, what fraction of variance is attributable to (a) encoder regularization,
27 (b) dynamics capacity, and (c) training protocol?

28 Our main finding is that the training protocol—specifically whether the model is trained with 1-step
29 teacher-forcing or 5-step BPTT—is the dominant factor. The regularizer matters but is secondary.
30 A surprising secondary finding is that 1-step-trained models show explosive A_k growth even on
31 a *stable linear* environment, indicating that the encoder geometry itself becomes unstable under
32 teacher-forcing regardless of regularization.

2 Background

Rollout amplification. We measure A_k via three complementary methods. The perturbation method computes:

$$A_k^{\text{pert}}(\varepsilon) = \mathbb{E}_{\mathbf{z}_0, \mathbf{u}} \left[\frac{\|f^k(\mathbf{z}_0 + \varepsilon \mathbf{u}) - f^k(\mathbf{z}_0)\|}{\varepsilon} \right] \quad (1)$$

where $\mathbf{u}$ is a random unit vector and $\varepsilon = 10^{-3}$. The Jacobian method computes the operator norm of the accumulated product Jacobian $\prod_{i=0}^{k-1} \partial f / \partial \mathbf{z} |_{\mathbf{z}_i}$. The trajectory-divergence method measures how the spread of nearby initial conditions evolves.

Marginal regularizers. SIGReg [Anonymous, 2025] pushes the empirical marginal $q(\mathbf{z})$ toward $\mathcal{N}(0, I)$ via a characteristic-function distance. We use a moment-matching proxy: $\mathcal{L}_{\text{reg}} = \|\boldsymbol{\mu}_{\text{batch}}\|^2 + \|\boldsymbol{\Sigma}_{\text{batch}} - I\|_F^2$.

3 Experimental Design

Factorial structure. We run a $2 \times 2 \times 2$ design: F1 (SIGReg on/off) $\times$ F2 (dynamics MLP depth: 1 vs. 3 hidden layers) $\times$ F3 (training: 1-step teacher-forcing vs. 5-step BPTT), giving 8 conditions.

Environments. (i) *Stable-linear*: $\mathbf{z}_{t+1} = A\mathbf{z}_t$, eigenvalues $\{0.7, 0.8, 0.9\}$, ground-truth $\lambda_{\max} < 0$. This is the sanity check: A_k must shrink. (ii) *2D spiral*: rotation + 5% expansion per step, $\lambda_{\max} \approx +0.049$. (iii) *Lorenz*: $\sigma=10, \rho=28, \beta=8/3, dt=0.01, \lambda_{\max} \approx +0.906$.

For each environment, 50,000 trajectories of length 100 are generated (80/10/10 split). 5 seeds per condition, 150 total runs. All models train for 20,000 steps, batch size 8,192, AdamW (lr = 3×10^{-4} , wd = 10^{-4}). Latent dimension $d_z = 16$. Encoder: 2-layer MLP (width 256, GELU, LayerNorm).

Side sweep. Latent dimension $d_z \in \{2, 8, 32\}$ on the 2D spiral (SIGReg + small + 1-step).

Oracle baseline. Dynamics trained directly on ground-truth states (large + 5-step BPTT).

Deviations from original brief. (1) n_steps reduced 50k $\rightarrow$ 20k (user-approved, matched compute across all conditions). (2) Batch size increased 256 $\rightarrow$ 8192 for GPU utilization (user-approved; larger batches improve SIGReg’s marginal estimate). (3) Jacobian method computed only at $k \in \{1, 5, 10\}$ with $n = 20$ samples (user-approved; large- k Jacobian products are numerically near-singular and covered by perturbation/traj_div). (4) Moment-matching SIGReg proxy used instead of original characteristic-function implementation.

4 Results

4.1 Variance Decomposition

Table 1 shows the variance decomposition of $\log A_{50}^{\text{pert}}$ per environment.

Table 1: Variance decomposition of $\log A_{50}^{\text{pert}}$ (η^2 , fraction of total variance). F1=SIGReg, F2=dynamics capacity, F3=training protocol.

Environment	F1	F2	F3	F1 $\times$ F2	F1 $\times$ F3	F2 $\times$ F3	Residual
Stable-linear	0.126	0.052	0.636	0.035	0.039	0.027	0.085
Spiral-2D	0.106	0.251	0.127	0.132	0.014	0.033	0.337
Lorenz	0.277	0.024	0.437	0.015	0.138	0.004	0.105

F3 (training protocol) is the dominant factor on stable-linear ($\eta^2 = 0.636$) and Lorenz ($\eta^2 = 0.437$). On spiral-2D, F2 (dynamics capacity) leads ($\eta^2 = 0.251$), with F3 and F1 comparable. F1 (SIGReg) is consistently secondary, with its largest effect on Lorenz ($\eta^2 = 0.277$).

Per the pre-registered decision criteria: $F3 \geq 2 \times F1$ on stable-linear and Lorenz, so *the original framing should be redirected toward training protocols*.

67 4.2 Sanity Check Failure

68 **Critical finding:** 17 of 40 stable-linear conditions exhibit $A_{50}^{\text{pert}} > 1$. For 1-step-trained mod-
69 els, A_{50} reaches values of 131 (NoReg_Small_1step), 952 (SIGReg_Small_1step), and 266 (SI-
70 GReg_Large_1step) on average across seeds. This is impossible if the latent dynamics faithfully
71 represent the underlying linear system.

72 The explanation lies in the encoder geometry: 1-step teacher-forcing trains the encoder and dynamics
73 jointly but only ever supervises single-step transitions. The encoder is free to warp the latent space in
74 ways that preserve 1-step prediction accuracy while creating geometrically unstable structure (e.g.,
75 expanding directions that cancel over one step but compound over many steps). SIGReg enforces a
76 Gaussian marginal, which forces the encoder to use more of the latent space (effective $d_z \approx 9$ –10 vs.
77 2–5 for NoReg conditions; see Appendix A). A fuller latent space under 1-step training creates more
78 opportunity for geometric instability.

79 5-step BPTT eliminates this problem: $A_{50} \leq 0.003$ for NoReg_Large_5step and ≤ 0.082 for
80 SIGReg_Large_5step on stable-linear—both well below 1.

81 4.3 Method Agreement

82 Perturbation and trajectory-divergence methods agree closely (within 20% for most conditions).
83 The Jacobian method gives higher values at $k \leq 10$ (computed via exact Jacobian products) and is
84 consistent in direction with perturbation. The Jacobian method is not computed at $k > 10$ (flagged
85 deviation; numerically near-singular).

86 4.4 Latent Width Sweep

87 On the 2D spiral (SIGReg + small + 1-step), A_{50}^{pert} : $d_z=2$: 0.60 ± 0.16 ; $d_z=8$: 1.19 ± 0.37 ; $d_z=32$:
88 7.29 ± 4.89 . Larger latent dimensions produce higher drift under 1-step teacher-forcing, consistent
89 with the encoder-geometry hypothesis: more dimensions provide more room for SIGReg to spread
90 the representation, creating more unstable directions.

91 4.5 Diagnostics

92 SIGReg successfully drives marginal KL to near-zero (1–53 nats vs. 37–969 for NoReg). Effective
93 latent dimension under SIGReg is 9–10 out of 16; under NoReg, 2–5. Val MSE is consistently low
94 across conditions (≤ 0.05), confirming models are not underfitting.

95 5 Discussion

96 The central result is that 5-step BPTT eliminates rollout drift even without regularization, while 1-step
97 teacher-forcing creates explosive drift even on stable systems. This redirects the research question:
98 rather than designing better marginal regularizers, the field should investigate *multi-step training*
99 *objectives* and their interaction with encoder geometry.

100 The SIGReg effect on Lorenz ($\eta^2 = 0.277$) suggests regularization matters more when the envi-
101 ronment is chaotic—possibly because the Gaussian prior provides an inductive bias that resists the
102 encoder’s tendency to overfit to local chaotic structure. This interaction between regularization and
103 environment complexity is a fruitful direction.

104 **Limitations.** Synthetic environments only; no image encoders; no action conditioning; 20k training
105 steps; moment-matching SIGReg proxy; A_k measures local sensitivity only. See Appendix B for full
106 list.

107 6 Conclusion

108 We asked: what causes rollout drift in latent world models? The answer, from a controlled causal
109 attribution experiment, is primarily the training protocol. Multi-step BPTT supervision forces the
110 encoder to learn geometrically stable latent representations; single-step teacher-forcing does not.

111 Marginal regularizers are secondary—important but not the root cause. We recommend that future
 112 work on latent world models evaluate training objectives before regularizer design.

113 **Acknowledgements.** This paper was assisted by Claude (Anthropic) for experiment execution and
 114 writing.

115 References

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121 A Full Results Tables

122 A.1 Master Results Table (Perturbation Method)

Table 2: A_k^{pert} mean $\pm$ 95% CI across 5 seeds. Selected columns ($k = 1, 10, 50$).

Env	Condition	A_1	CI	A_{10}	CI	A_{50}	CI
Stable-linear	NoReg_Large_1step	0.742	0.028	0.949	0.352	0.846	1.160
	NoReg_Large_5step	0.528	0.029	0.171	0.019	0.003	0.000
	NoReg_Small_1step	0.803	0.027	2.542	1.031	131.937	152.188
	NoReg_Small_5step	0.585	0.013	0.241	0.030	0.037	0.030
	SIGReg_Large_1step	1.163	0.016	5.160	0.755	266.242	358.819
	SIGReg_Large_5step	0.914	0.015	1.524	0.346	0.055	0.020
	SIGReg_Small_1step	1.174	0.019	6.038	2.366	952.419	873.118
	SIGReg_Small_5step	1.033	0.010	1.653	0.378	0.028	0.019
Spiral-2D	NoReg_Large_1step	0.721	0.072	1.033	0.231	1.255	0.702
	NoReg_Large_5step	0.549	0.026	0.672	0.180	0.875	0.538
	NoReg_Small_1step	0.745	0.076	1.319	0.373	2.964	1.953
	NoReg_Small_5step	0.525	0.046	0.728	0.227	0.622	0.238
	SIGReg_Large_1step	0.767	0.015	0.615	0.025	0.942	0.125
	SIGReg_Large_5step	0.661	0.018	0.508	0.026	0.721	0.060
	SIGReg_Small_1step	0.890	0.012	1.125	0.095	6.916	4.625
	SIGReg_Small_5step	0.808	0.014	0.766	0.063	3.087	0.936
Lorenz	NoReg_Large_1step	0.955	0.050	1.814	0.183	37.349	31.300
	NoReg_Large_5step	0.587	0.022	0.606	0.054	1.523	0.381
	NoReg_Small_1step	0.937	0.044	2.023	0.444	34.220	44.032
	NoReg_Small_5step	0.537	0.022	0.649	0.079	2.270	1.591
	SIGReg_Large_1step	1.114	0.026	4.217	0.739	225.803	280.285
	SIGReg_Large_5step	0.822	0.018	1.040	0.078	3.143	1.166
	SIGReg_Small_1step	1.069	0.028	2.765	0.589	52.218	53.547
	SIGReg_Small_5step	0.706	0.017	0.867	0.112	4.094	2.325

123 B Caveats and Confounds

- 124 1. **Synthetic-only environments.** Results may not transfer to JEPA-family models trained on
 125 images.
- 126 2. **No image encoder.** SIGReg’s interaction with image encoder pathologies is not tested.
- 127 3. **No action conditioning.** Autonomous dynamics may differ from action-conditioned ones.
- 128 4. **Limited latent dimensions.** Behavior at $d_z = 256$ or 1024 (realistic for vision models) is
 129 not tested.

- 130 5. **Fixed-compute training (20k steps).** Findings about F1 may depend on convergence speed
131 differences between SIGReg and no-reg conditions.
- 132 6. **Moment-matching SIGReg.** $\mathcal{L} = \|\mu\|^2 + \|\Sigma - I\|_F^2$ was used instead of the original
133 characteristic-function implementation.
- 134 7. A_k **measures local sensitivity, not basin-of-attraction structure.**
- 135 8. **Encoder warping.** Ground-truth Lyapunov exponent of state space $\neq$ Lyapunov exponent
136 of learned latent space. Normalized A_k partially addresses this.
- 137 9. **Jacobian method limited to $k \leq 10$.** Large- k Jacobian products were excluded due to
138 numerical near-singularity.
- 139 10. **5 seeds per condition.** Variance estimates are coarse; some CIs are very wide (especially
140 on stable-linear 1-step conditions due to seed-to-seed variability in encoder geometry).

141 C Session Log

142 **Human inputs.** Topic: causal attribution of rollout drift in latent world models (via
143 preliminary_experiment_brief.md). Pre-registration: skipped (user decision). Approved devi-
144 ations: seeds 10 $\rightarrow$ 5; n_steps 50k $\rightarrow$ 20k; batch_size 256 $\rightarrow$ 8192; Jacobian limited to $k \leq 10$.

145 **AI contributions.** All code (environments, models, training loop, metrics, analysis, figures, report)
146 was written by Claude. Paper text was drafted by Claude. No Codex feedback was obtained (not
147 available on remote machine).

148 **Compute.** NVIDIA RTX A4000 (16GB), 150 runs, approximately 6 hours wall-clock.